

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of KarXT + KarX-EC for the Treatment of Cognitive Impairment Associated with Mild to Moderate Alzheimer's Disease (MINDSET 2) **Short Title/Acronym:** MINDSET 2 **Protocol Number:** CN012-0052 **NCT Number:** NCT06976203 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** KarXT (xanomeline/trospium combination) + KarX-EC **Phase of Development:** Phase 3 **Medical Monitor:** BMS Clinical Trials Contact Center (Email: clinical.trials@bms.com / Phone: 855-907-3286)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of KarXT + KarX-EC compared to placebo in treating cognitive impairment in patients with mild to moderate Alzheimer's Disease, assessed by the mean change from baseline in the Alzheimer's Disease Assessment Scale-Cognitive Subscale 11 (ADAS-Cog11) at Week 24. **Secondary Objectives:** To evaluate the effect on overall clinical and functional status using the Clinician's Interview-Based Impression Plus Caregiver Input (CIBIC+) at Week 24, and to evaluate the long-term safety and tolerability of the intervention.

2.2 Background & Rationale Current treatments for cognitive impairment in Alzheimer's disease often focus on acetylcholinesterase inhibitors or NMDA receptor antagonists and have limited efficacy or unfavorable side-effect profiles. KarXT + KarX-EC acts via a novel mechanism by modulating muscarinic receptors (M1/M4 muscarinic agonist), which has shown promise in improving memory, thinking, and behavioral symptoms (such as agitation) without the side effects typically associated with systemic muscarinic activation. This study aims to provide Phase 3 evidence to potentially offer a groundbreaking, more tolerable therapeutic option for AD symptom management.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel-Group (1:1 allocation to KarXT+KarX-EC or placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 586 participants **Number of Sites:** Approximately 123 locations globally **Estimated**

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 60 to 85 years. 2. Confirmed diagnosis of Alzheimer's disease (AD) at the mild (stage 4) or moderate (stage 5) dementia stages, as defined by NIA-AA criteria, with pathology confirmed via stepwise diagnostic approach (e.g., Amyloid PET or CSF lumbar puncture). 3. Mini-Mental State Examination (MMSE) score ranging from 12 through 22, inclusive, at screening. 4. Must have a designated caregiver who maintains adequate contact (~10 hours/week or more) and is willing to attend study visits and oversee compliance.

4.2 Exclusion Criteria 1. History of schizophrenia, other chronic psychosis, bipolar disorder, or severe psychiatric symptoms. 2. Previous exposure to KarXT or current treatment with disease-modifying anti-amyloid therapies for AD within the past 6 months prior to screening. 3. Significant or severe medical conditions compromising safety or study completion, including moderate-severe renal impairment ($\text{eGFR} < 50 \text{ mL/min}$), hepatic impairment, active biliary disease, urinary retention, or unstable hypertension/tachycardia.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Active study drug (KarXT + KarX-EC) or matching placebo administered daily. **Route of Administration:** Oral (capsule/pill form). **Duration of Treatment:** 24 weeks of double-blind treatment. (Note: The study features an optional Open Label Extension [OLE] period making total possible duration up to 57 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care medications (e.g., acetylcholinesterase inhibitors) provided they have been at a stable dose for at least 12 weeks prior to screening. **Prohibited:** Disease-modifying anti-amyloid therapies for AD within 6 months prior to screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Weeks -5 to 0	Informed Consent, MMSE, MRI, amyloid PET or lumbar puncture, vital signs, ECG
Baseline	Day 0	Randomization, First Dose, Caregiver protocol education
Interim	Up to Week 24	Efficacy Assessment (ADAS-Cog11, CIBIC+), Safety Labs, Caregiver reporting
End of Study	Week 28	Final Double-Blind Vital Signs, 4-week post-

treatment follow-up, or transition to Open Label Extension (OLE)
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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in the Alzheimer's Disease Assessment Scale-Cognitive Subscale 11 (ADAS-Cog11) at Week 24. **Measurement Method:** Clinician-administered standardized cognitive assessment.

7.2 Secondary & Exploratory Endpoints 1. Clinician's Interview-Based Impression Plus Caregiver Input (CIBIC+) at Week 24. 2. Long-term safety, overall function, and tolerability (assessed further through the optional OLE phase).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient and caregiver reporting, physiological assessments (ECG, vitals), and lab collections. **Serious Adverse Events (SAEs):** Safety monitoring includes monitoring for peripheral anticholinergic/cholinergic effects (mitigated by trospium component); reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety monitoring framework coordinated by Bristol-Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy measures (all randomized subjects); Safety Set for all participants who received at least one dose. **Sample Size Justification:** Target N=586, sufficiently powered to detect a statistically significant difference in ADAS-Cog11 scores between the KarXT + KarX-EC and placebo arms. **Handling of Missing Data:** Mixed-Model for Repeated Measures (MMRM) or Multiple Imputation (MI) used for primary continuous endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of trial sites to ensure protocol and caregiver compliance. **Audit Trail:** Secure eCRF entry locking, central review of blinded cognitive scale scoring (ADAS-Cog11), and standard data clarification procedures.

11. Study Identifiers & Governance

Primary ID: CN012-0052 **Registry Number:** NCT06976203 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** MINDSET 2
Official Regulatory Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study to Evaluate the Efficacy and Safety of KarXT + KarX-EC for the Treatment of Cognitive Impairment Associated with Mild to Moderate Alzheimer's Disease (MINDSET 2)

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Alzheimer's Disease (Mild to Moderate Dementia Stages) **Diagnosis Threshold:** NIA-AA core clinical criteria (stage 4/5); MMSE 12-22; confirmed AD pathology. **Medical Methodology:** Muscarinic receptor modulation (M1/M4 agonism via xanomeline, peripheral antagonism via trospium). **Optimization Target:** Improvement of cognitive impairment and mitigation of behavioral decline.

13. Study Procedures & Methodology

Management Pathway: Standard Alzheimer's clinical trial assessments including structural imaging (MRI) and biomarker confirmation. **Education Protocol (HCP):** Training on ADAS-Cog11 and CIBIC+ administration, caregiver interview standardization. **Education Protocol (Patient/Caregiver):** Caregiver instructions on overseeing medication adherence (~10 hours/week minimum contact), recognizing AEs, and recording observations. **Quality Audit Mechanism:** Centralized rater review for cognitive assessments and continuous caregiver input tracking.

14. Observation & Follow-Up Schedule

Total Duration: 33 weeks for the double-blind phase (5-week screening + 24-week treatment + 4-week follow-up), with an optional OLE pushing total expected duration up to 57 weeks. **Onsite Visit Cadence:** Baseline, regular intervals up to Week 24, and Week 28. **Remote Monitoring Frequency:** Routine caregiver check-ins as defined by site protocols. **Checklist Review Interval:** Regular safety parameter checks up to the 24-week efficacy mark.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |